00404

Sri Lanka Institute of Information Technology

B.Sc. Honours Degree in Information Technology

Specialized in Information Technology

Final Examination
Year 4, Semester 2 (2022)

IT4090 – Cloud Computing

Duration: 2 Hours

November 2022

Instructions to Candidates:

- This paper is preceded by 10 minutes reading period. The supervisor will indicate when answering may commence.
- This paper has 4 questions.
- Answer all questions in the booklet given.
- The total mark for the paper is 100.
- This paper contains 4 pages, including the cover page.
- Electronic devices capable of storing and retrieving text, including calculators and mobile phones are not allowed.

Question 1 (20 marks)

ABC is a startup company provides different IT based solutions. Based on requirements of a client, they have implemented a system which sells tickets to concerts, sports events, and music gigs. Tickets are sold offline through ticket counters and online through a website called TicketOnline.

Customers visiting TicketOnline can browse list of available shows, find out more information on each show, and finally purchase tickets online. When a ticket is purchased, it's reserved but will not be processed immediately. Other processes such as generating ticket and sending the generated ticket along with the receipt will be done asynchronously in a few minutes time.

During peak season, TicketOnline suffered from heavy traffic that caused slow response time. Additionally, the current infrastructure setup is not designed to be highly available. This results in several periods of downtime each year. These are current challenges faced by TicketOnline.

While using the existing system clients have faced some issues and they have informed ABC company about this matter. ABC's IT manager recognizes the issues and decides to make some improvement to bring better competitive advantages to the company. When researching, he discovered that Cloud computing may be a good solution to address the issues. Another option would be to purchase a more powerful set of hardware that could handle the load.

You are hired as an Infrastructure Engineer to find a solution for this scenario.

a) Based on the above scenario, explain three (3) benefits that the system would gain by migrating to a Cloud Service Provider. (5 marks)

b) Explain three (3) disadvantages that the system would gain by purchasing more powerful hardware components. (5 marks)

c) Sketch how you are going to achieve high availability in your system with aid of a diagram. You can assume the application is in 3 tier architecture). (5 marks)

d) IT Manager state that "***being On-Premises is better than migrating to Cloud in this scenario***". As the infrastructure Engineer do you agree with this? Justify your answer. (5 marks)

Question 2 (25 marks)

a) Explain the Principle of Least Privilege (PLOP). (3 marks)

b) IAM is a framework of policies and technologies to ensure users/process have the appropriate access to technology resources. Discuss three (3) IAM components. (5 marks)

c) Each time an enterprise deploys a new application, end users must create a new set of credentials to remember. This results of having too many passwords. Some users reuse the same two to four passwords for various accounts, and just one password for all their applications. This type of weak password hygiene makes easier for hackers to use stolen credentials to access other critical data compromising individuals and businesses. Therefore, organizations need to be able to provide users with easy access to all of their applications by adopting tools like **Federated Identity Management (FIM)** and **Single Sign-On (SSO).**

   i. How is FIM different from SSO? (3 marks)

   ii. Apply two (2) ideal use cases for each approach. (4 marks)

d) Differentiate Symmetric and Asymmetric encryption with aid of diagrams. (5 marks)

e) With the adoption of the cloud comes the need to ensure the organization's cloud security strategy that can protect against the top threats to cloud security. Do you agree with this? Justify your answer. (5 marks)

Question 3 (30 marks)

a) What is Microservices architecture? Briefly discuss in your own words. (3 marks)

b) State three (3) characteristics of Microservices and briefly explain each. (3 marks)

c) Compare and contrast Microservices architecture with Monolithic Architecture, with at least three (3) points. (6 marks)

d) A new startup company is thinking of developing a new web application for e-commerce platforms. They have a requirement to release their initial version as soon as possible to the market because they have one client already waiting for their product. You, as a senior development team member have been asked the question, if they should go with microservices approach for the development of the application now itself, or should they go with Monolithic architecture for now, and refactor their architecture to Microservices later as the product evolves? Evaluate the above problem and justify your position. (7 marks)

e) If the above software is to be designed with a microservices approach, it could either be developed and deployed using either a Container based approach or using a Functions as a Service (FaaS) approach. Compare and contrast these approaches and recommend a suitable approach to follow given the urgency to release the product. Justify why the selected approach is the better approach in the given context.

(7 marks)

f) Is Functions as a Service (FaaS) always the preferable choice when it comes to development and deployment of microservices? Briefly discuss. (4 marks)

Question 4 (25 marks)

a) Describe the three main barriers for IT Innovation. (3 marks)

b) Well-Architected helps cloud architects build secure, high-performing, resilient, and efficient infrastructure for a variety of applications and workloads.

" Organizations can ensure Cloud native applications are architected in a way that aligns with business needs, are highly performant and reliable, and are cost-effective."

Do you agree with this statement? Justify your answer. (4 marks)

c) Evaluate AWS Well Architected Framework and Azure Well Architected Framework. (6 marks)

d) Most organizations today are already working in multi-cloud environments. Many businesses experience challenges when it comes to integrating data sources and applications within a multi-cloud infrastructure.

i. Define the concept of Multi cloud integration. (3 marks)

ii. Mention three (3) advantages of multi cloud advantages. (3 marks)

iii. To maximize the value of a multi-cloud environment, it's important to mitigate common challenges that often arise with multi-cloud integration. Discuss three (3) challenges. (6 marks)

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